

Shiladitya De

Curriculum Vitae

Date of Birth : August 19th, 2001
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EDUCATION

- **Integrated Bachelor and Master of Technology (Dual Degree)** 2020-2025
Computer Science and Engineering *Language of Instruction: English*
Indian Institute of Technology (IIT) Kharagpur, Kharagpur, India
Cumulative GPA (CGPA): **9.56/10** *Department Rank: 4 out of 70*
- **Higher Secondary Education** 2018-2020
Kendriya Vidyalaya IIT Kharagpur, Kharagpur, India *Language of Instruction: English*
Percentage: **97%** *Rank: 1 out of 65 in school*
- **Secondary Education** 2013-2018
Kendriya Vidyalaya IIT Kharagpur, Kharagpur, India *Language of Instruction: English*
Percentage: **95.8%** *Rank: 10 out of 160 in school*

RESEARCH INTERESTS

My research experience lies in **usable security and privacy**, with a focus on understanding security and privacy challenges in real-world systems and user-facing technologies. I have worked on problems spanning **software security**, **security issue reporting**, and **user decision-making in digital payment systems**. I am interested in pursuing research broadly in **human-centric security and privacy**, including problems at the intersection of users, software, and security.

PUBLICATIONS

- Nandini Bajaj*, Shiladitya De*, Kshitiz Sharma, Xinru Page, Bart Knijnenburg, Mainack Mondal. “To Pay or Not to Pay?": Understanding User Decision-Making and Influence of Nudges in UPI Apps, *ACM Asia Conference on Computer and Communications Security (ASIA CCS '26)*. <https://doi.org/10.1145/3779208.3785285>
* Equal contribution
- Rajdeep Ghosh, Shiladitya De, and Mainack Mondal. “I wasn’t sure if this is indeed a security risk”: Data-driven Understanding of Security Issue Reporting in GitHub Repositories of Open Source npm Packages, *34th USENIX Security Symposium (USENIX Security'25)*. <https://dl.acm.org/doi/10.5555/3766078.3766189>

WORK EXPERIENCE

- **Member Technical,** June 2025 – Present
D. E. Shaw India Private Limited, Hyderabad, India
 - Developed financial analytics infrastructure and algorithms for **derivatives pricing**, **credit spreads**, **realized volatility**, and **bond yield computation**.

INTERNSHIPS

- **Revamping the Taylor Series Framework for Optimized Real-Time Analytics**
D. E. Shaw India Private Limited, Hyderabad, India May 2024 – July 2024
 - Redesigned the Taylor Series framework using **matrix-vector multiplications**, improving extensibility and real-time performance while maintaining **backward compatibility**.

SELECTED PROJECTS

- **Alice in Trackerland: Exploring Real-World Utilities of Continuous Location Tracking Apps,**
Masters Thesis Project August 2024 – April 2025
Guides: Mainack Mondal & Satrajit Ghosh
- **PaPAd: Practical and Private Advertising,**
Bachelors Thesis Project August 2023 – April 2024
Guides: Mainack Mondal & Satrajit Ghosh
- **Scalable Distributed Database,**
Course Project, Distributed Systems February 2024 – April 2024
Instructor: Sandip Chakraborty
 - Built a **scalable distributed database** in Go with **range-based sharding**, WAL, and a Dockerized **load-balanced** server architecture
- **Query-by-Example for Scientific Article Retrieval,**
Course Project, Information Retrieval February 2023 – April 2023
Instructor: Somak Aditya
 - Built a **Query-by-Example** system for faceted scientific article retrieval using an ensemble of **6 language models**, improving NDCG by **7.97%**

AWARDS AND ACHIEVEMENTS

- All India Rank **3402** in *JEE Advanced, 2020* amongst ~150,000 shortlisted candidates
- All India Rank **4661** in *JEE Main, 2020* amongst ~1,000,000 candidates
- Rank **58** in *West Bengal Joint Entrance Examination 2020* amongst ~90,000 candidates
- Secured **second** position in **Maths Olympiad General Championship** conducted by Technology Students Gymkhana, IIT Kharagpur in January 2024
- Qualified for the **Indian National Astronomy Olympiad (INAO)** in **2019**
- Qualified for **Junior Maths Olympiad (JMO)** and **Regional Mathematics Olympiad (RMO)** in **2016** and **2017** respectively

COURSEWORK INFORMATION

- **Computer Security:** Usable Security and Privacy, Information and System Security, Cryptography and Network Security, Foundations of Cryptography, Cryptographic Protocol Theory, Computational Number Theory
- **Machine Learning:** Machine Learning, Natural Language Processing, Linear Algebra for AI and ML, Statistical Foundations of AI and ML, Information Retrieval
- **Other courses in Computer Science:** Switching Circuits & Logic Design*, Compilers*, Systems Programming Laboratory, Computer Networks*, Operating Systems*, Programming & Data Structures*, Computer Organisation & Architecture*, High Performance Computer Architecture, Software Engineering*, Database Management Systems*, Formal Language & Automata Theory, Algorithms-I*, Discrete Structures, Algorithms II, Advances in Operating Systems Design, Distributed Systems, Parallel Algorithms, Advanced Graph Theory
- **Mathematics:** Advanced Calculus, Linear Algebra, Numerical & Complex Analysis, Probability & Statistics

* marked courses include laboratory component as well

OTHER ACTIVITIES, ACADEMIC DUTIES & POSITIONS OF RESPONSIBILITY HELD

- **Co-designed** and served as **teaching assistant**, *Design Optimization of Computing Systems*, CSE, IIT KHARAGPUR, Autumn 2024
- **Teaching Assistant**, *Information and System Security*, CSE, IIT KHARAGPUR, Spring 2025
- **Presenter**, *ACM Asia Conference on Computer and Communications Security (ASIACCS) 2026*
- **Attendee**, *Theory and Practice of Multi-Party Computation (TPMPC) 2025*, IISc Bangalore
- **Department Representative** to the **Undergraduate Council** for the session 2022-23
- Conducted discussion and star-gazing sessions as a member at Space Technology Student Society
- Participated in various culinary competitions organized by AROMA, IIT Kharagpur